

DESIGN PROPOSAL

Photonic Mach-Zehnder Interferometers

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Abstract: This draft proposes three unbalanced Mach-Zehnder interferometers (MZIs) on the 220 nm silicon-on-insulator platform, with the goal of extracting the group index n_g of a 500 nm $\times$ 220 nm strip waveguide for TE polarization. It include three MZI , three beam splitters, and three Y-branches. Path difference varies from 30 to 400 μm and designed for TE polarization at 1550 nm.

1. Introduction

In this draft, the layout contains three MZIs based on the circuit of the course example: an input grating coupler, a Y-branch splitter, two arms of different length, a broadband directional coupler (BDC) that recombines the light, and two output grating couplers. In the final layout, the three MZIs will be three different designs, so that their results can be compared: The path varies between 50 to 238 nm. The free spectral range (FSR) of an unbalanced MZI depends on the group index n_g of the waveguide. Measuring the FSR therefore gives n_g , which can be compared with the simulated value. The design objectives are:

- to design unbalanced MZIs whose FSR is smaller than the measurement bandwidth (about 50 nm), so that several fringes are visible.
- to extract n_g from the 500 x 220 nm TE strip waveguide and compare it with simulation
To study how the FSR systematically changes across the three different path-length differences (ΔL).

2. Theory

The basic equations used in this proposal are summarized below.

Group index (from the effective index)

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda} \quad (a)$$

MZI output intensity (ideal splitters, no loss)

$$I_{out} = \frac{I_{in}}{2} [1 + \cos(\beta \Delta L)], \quad \beta = \frac{2\pi n_{eff}}{\lambda} \quad (b)$$

Here $\Delta L = L_2 - L_1$ is the length difference between the two arms. With a BDC as the combiner, the MZI has two complementary outputs: when one is at a maximum, the other is at a minimum.

Free spectral range (spacing between two adjacent minima)

$$FSR \approx \frac{\lambda^2}{n_g \Delta L} \quad (c)$$

Group index extracted from the FSR

$$n_g \approx \frac{\lambda^2}{\Delta L \cdot FSR} \quad (d)$$

3. Modelling and simulation

3.1. Waveguide geometry and mode profile

The waveguide is a silicon strip of $500 \text{ nm} \times 220 \text{ nm}$ with SiO_2 cladding, designed for TE polarization at 1550 nm . The mode was calculated with Lumerical MODE (FDE solver).

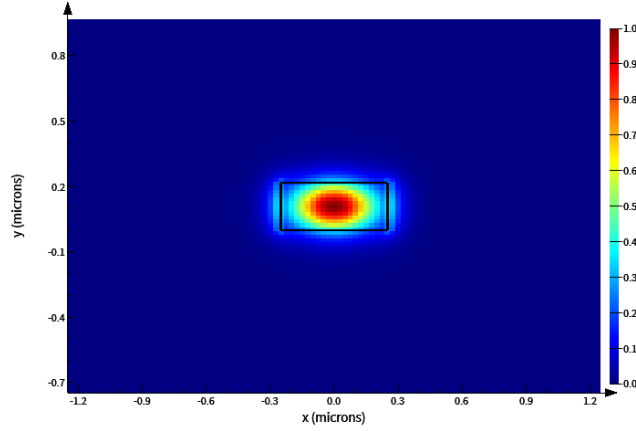


Figure 1: Mode profile from Lumerical MODE (linear scale)

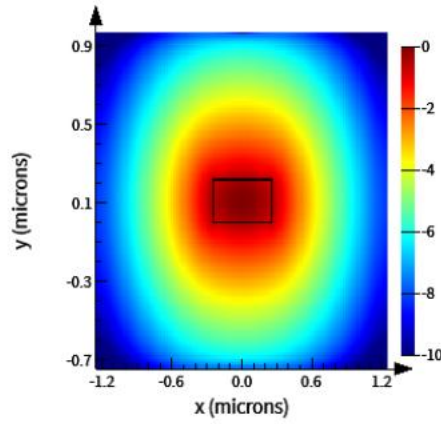


Figure 2: Mode profile from Lumerical MODE (log scale)

3.2. Effective and group index

A wavelength sweep from 1500 to 1600 nm gives the effective and group index shown in Fig. 3 and Fig 4 respectively at 1550 nm where $n_{\text{eff}} = 2.502$ and $n_g = 4.194$.

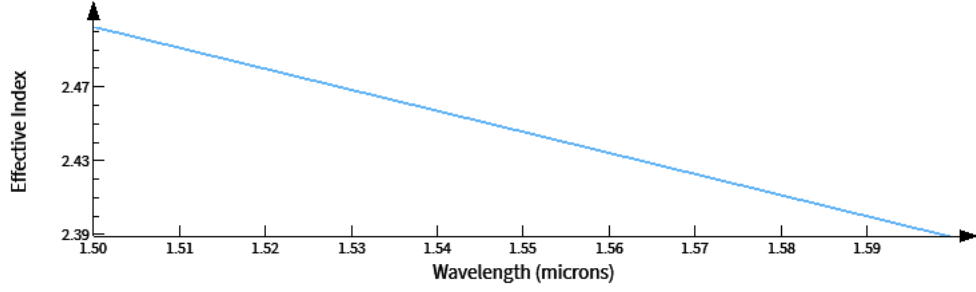


Figure 3: Effective index of TE_0 mode vs wavelength from Lumerical MODE.

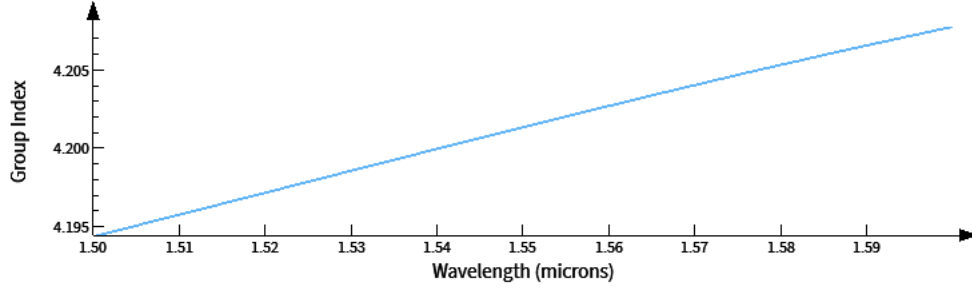


Figure 4: Group index of TE_0 mode vs wavelength from Lumerical MODE.

3.3. Waveguide compact model

For the circuit simulation, the effective index is fitted with a second-order polynomial around $\lambda_0 = 1.55 \mu\text{m}$:

Compact model of the effective index

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$$

with $n_1 = 2.445$, $n_2 = -1.129 \mu\text{m}^{-1}$ and $n_3 = 0.1250 \mu\text{m}^{-2}$ (λ in μm).

3.4. Circuit design

The layout (EBeam_fahmida) was implemented in KLayout using the SiEPIC EBeam component library. It features three unbalanced MZI structures connected to shared input/output optical grating couplers optimized for 1550 nm TE polarization. The waveguides consist of 500 nm wide by 220 nm thick silicon strip profiles.

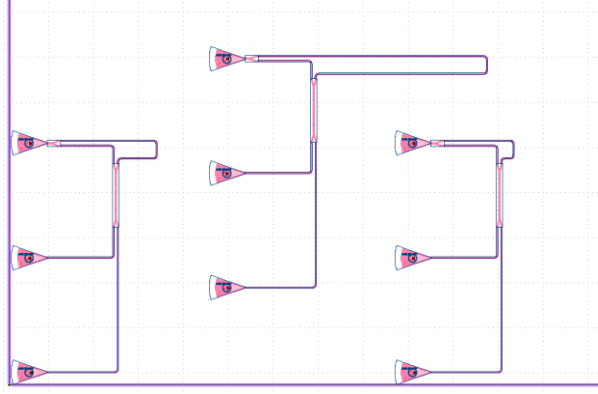


Figure 5: Schematic of the MZI circuit in KLayout

3.5. Parameter variations

Table 1 lists the three devices and the expected FSR, calculated with equation (c) and $n_g = 4.194$ [value from MODE]. In the draft layout the MZIs are based on the course example; in the final layout they will be three different designs.

$$FSR = \frac{\lambda_0^2}{n_g \cdot \Delta L}$$

Table 1: Proposed devices and expected FSR

<i>Device</i>	<i>ΔL (μm)</i>	<i>Expected FSR (nm)</i>
<i>MZI1</i>	93.137	6.15
<i>MZI2</i>	388.397	1.49
<i>MZI3</i>	33.817	16.93

3.6. Simulated transmission spectra

Each MZI was simulated in Lumerical INTERCONNECT from 1500 to 1600 nm, using the compact model of section 3.3. The simulated spectra display high-contrast periodic interference fringes superimposed on the broad wavelength-dependent transmission envelope of the grating couplers. Across the 100 nm bandwidth, MZI1 exhibits approximately 16 periodic fringes, MZI2 shows 69 fringes, and MZI3 displays 5 fringes, confirming that larger path-length differences (ΔL) yield tighter fringe spacing (smaller FSR) and a higher density of interference ripples.

Each MZI was simulated in Lumerical INTERCONNECT from 1500 to 1600 nm, using the compact model of section 3.3.

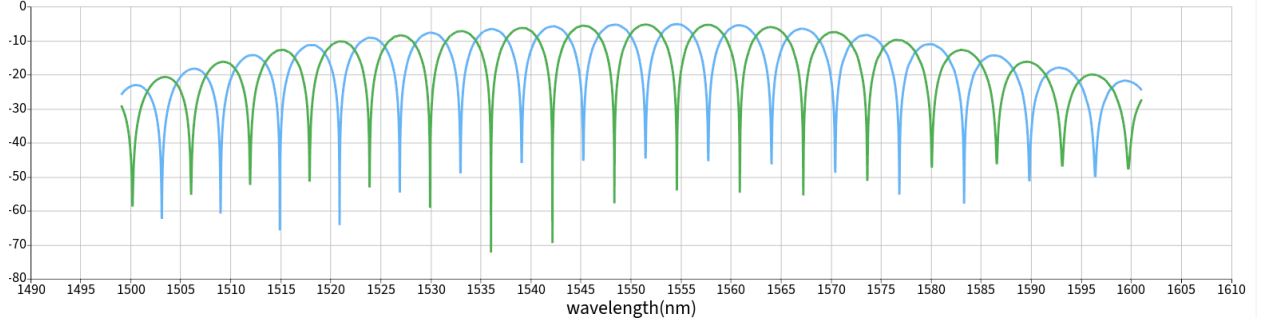


Figure 6 :Transmission spectra of the MZI1 from Lumerical INTERCONNECT

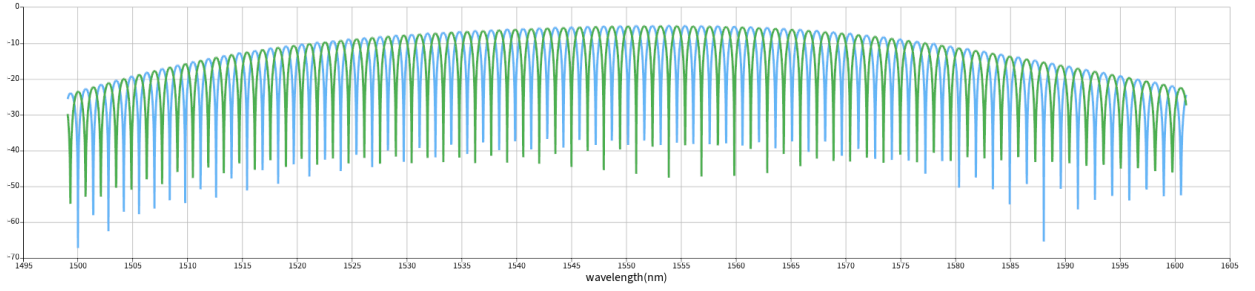


Figure 7 : Transmission spectra of the MZI2 from Lumerical INTERCONNECT

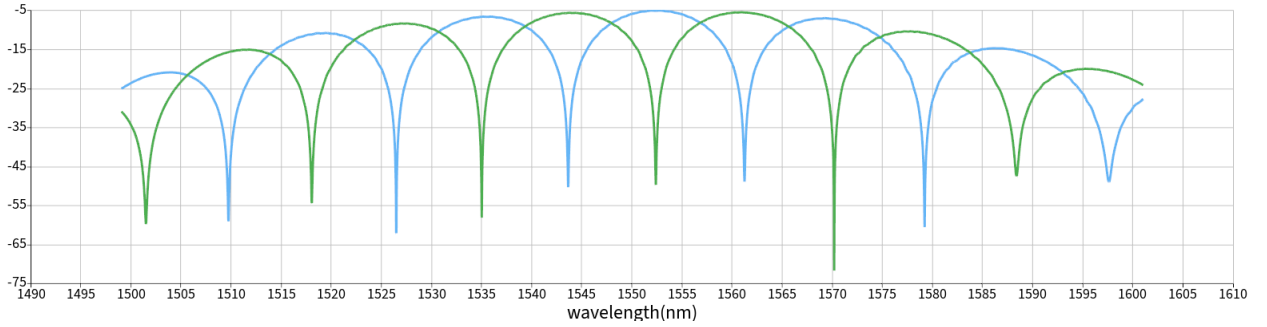


Figure 8: Transmission spectra of the MZI3 from Lumerical INTERCONNECT

3.7. Test of the extraction equation

To check equation (d), the FSR is read from the simulated spectra and used to calculate n_g , which is then compared with the value from MODE (Table 2).

Table 2: Group index extracted from the INTERCONNECT spectra (near 1550 nm)

Device	FSR from simulation (nm)	FSR expected	n_g from equation (d)	n_g from MODE
MZI1	6.26	6.15	4.194	4.194
MZI2	1.47	1.49	4.194	4.194
MZI3	17.36	16.93	4.194	4.194

4. Fabrication

To be completed in the final layout phase.

5. Experimental data

To be completed when the measurements are available.

6. Analysis

To be completed in the analysis phase.

7. Conclusion

To be completed in the final report.

References

1. L. Chrostowski and M. Hochberg, Silicon Photonics Design: From Devices to Systems (Cambridge University Press, 2015).